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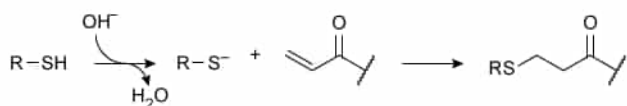


Figure 1 Reaction of a thiol with an acrylamide

Experiment 1

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Experiment 2

A fluorescent amino acid was added to Ras_{G12C} at a position far away from the ARS-853 binding site. The change in fluorescence intensity over time upon addition of ARS-853 was measured. Two time constants were calculated from the fluorescent decay signal, consistent with a two-step reaction mechanism of binding followed by covalent attack.

RASAL is a protein that regulates Ras activity. When RASAL binds Ras at an

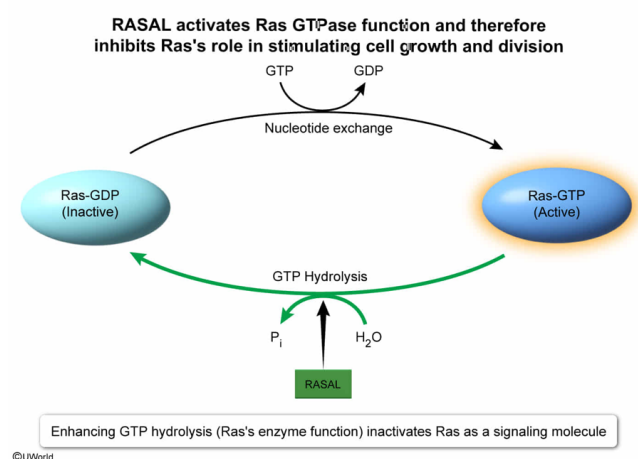
allosteric site, it lowers the activation energy of the GTP hydrolysis reaction that Ras catalyzes. RASAL is best described as:

- ☐ A. an activator of both Ras's GTP hydrolysis function and of cell growth and division. [39%]
- ☐ B. an inhibitor of both Ras's GTP hydrolysis function and of cell growth and division. [6%]
- ✓ ☒ C. an activator of Ras's GTP hydrolysis function but an inhibitor of cell growth and division. [50%]
- ☐ D. an inhibitor of Ras's GTP hydrolysis function but an activator of cell growth and division. [3%]

Correct

49% answered correctly

Explanation:



Proteins can be **allosterically** activated or inactivated by specific **ligands** that they bind. Regulatory molecules that affect the ligands will further alter the protein's function.

G proteins are common signaling proteins in cells that are switched "on" when they bind GTP. After the G protein hydrolyzes GTP into GDP, the G protein **switches "off"** until the GDP is exchanged for a new GTP. Guanine nucleotide exchange factors (GEFs) catalyze this exchange and therefore stimulate G protein signaling activity. Ligand-bound **G protein–coupled receptors (GPCRs)** serve as GEFs for their associated heterotrimeric G proteins, while small, cytosolic G proteins have small, soluble GEFs.

In contrast, molecules called GTPase-activating proteins (GAPs) can inhibit G protein signaling activity by enhancing their intrinsic hydrolase activity. Because

enzymatic function is enhanced, the active (GTP-bound) form of the G protein has a **shorter lifetime** because the GTP itself has a shorter lifetime.

The passage states that Ras is a G protein that, when bound to GTP, activates Raf, triggering a **phosphorylation cascade** leading to cell growth and division. The question states that RASAL binding to Ras lowers the energy of the the activation energy of the hydrolysis reaction. Because enzyme activity is tied to the **lowering of activation energy**, RASAL serves as a GAP that *enhances* Ras's *GTP hydrolysis function* and *inhibits its signaling function* by shortening the lifetime of the Ras-GTP complex.

(Choice A) Because GTP hydrolysis function is enhanced, GTP is degraded more quickly. GTP is required for Ras to function as a signaling molecule; therefore, cell growth and division are *inhibited* by RASAL.

(Choices B and D) Enzymes function by lowering the activation energy of the reaction they catalyze. Because RASAL causes the activation energy to lower even further, RASAL would *stimulate*, not inhibit, Ras's GTP hydrolysis function.

Educational objective:

Proteins can be allosterically regulated by their ligands. If a ligand stimulates protein function, then degradation of that ligand inhibits protein function.

Time spent:0 sec

Subject:
Biochemistry

QID:404174

System:
Amino Acids and Proteins

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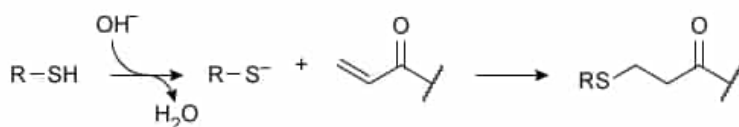


Figure 1 Reaction of a thiol with an acrylamide

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The mutation G12C results in:

- ☐ A. an achiral residue being replaced with a residue that has an *S* absolute configuration. [37%]
- ☐ B. a residue with no *R/S* designation being replaced with a residue that has a *D* configuration. [16%]
- ☐ C. a residue with an *S* absolute configuration being replaced with a residue that has an *L* configuration. [10%]
- ✓ ☒ D. a residue with no *D/L* designation being replaced with a residue that has an *R* absolute configuration. [35%]

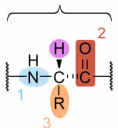
Correct

36% answered correctly

Explanation:

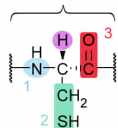
Glycine is an achiral amino acid; cysteine is a chiral, (*R*)-configuration, L-amino acid

All residues except Gly and Cys



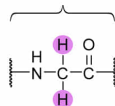
- The α -carbon is chiral (four distinct substituents)
- Chiral amino acids are naturally L-configuration
- Most chiral amino acids are (S)-configuration

Cysteine residue



- The α -carbon is chiral (four distinct substituents)
- Chiral amino acids are naturally L-configuration
- Cysteine is the only (*R*)-configuration amino acid

Glycine residue



- The α -carbon is bonded to two H atoms and is therefore achiral

Amino acids are the building blocks of proteins and consist of an α -carbon **participating in four bonds**. The four substituents the α -carbon is bonded to, in order of typical **stereochemical priority**, are:

1. An N atom, in the form of either an **amine group** (in free amino acids or at the **N-terminus** of a peptide) or an **amide group** (as part of the **peptide bond** to another amino acid)
2. A **carbonyl carbon**, in the form of either a **carboxylic acid** (in free amino acids or at the C-terminus of a peptide) or an amide group (as part of the peptide bond to another amino acid)
3. A variable R group that constitutes the amino acid's **side chain**
4. A hydrogen atom

By convention, biochemical molecules are assigned an L or D stereochemical designation based on their **stereocenter's** similarity to L- or D-**glyceraldehyde**. **Natural amino acids are typically L configuration**. Given priority rankings of N atom > carbonyl carbon > side chain > α -hydrogen, the **absolute stereochemistry of amino acids is typically S**.

There are **two exceptions** to the L,S configuration of the natural amino acids: **glycine** and **cysteine**.

- Because the side chain of glycine is a hydrogen atom, the α -carbon of glycine is bonded to **two identical substituents** and is **achiral**. Therefore, free glycine amino acids and glycine residues in peptides have **no stereochemical designation**.
- Because sulfur has a higher **atomic number** than oxygen, the cysteine side chain takes a **higher priority** than the carbonyl carbon. Although natural cysteine still has an L configuration, its **absolute stereochemical designation is R** rather than S.

The G12C mutation replaces the glycine at position 12 with cysteine. Therefore, it replaces an achiral residue (glycine), which does not have a D or L designation, with an amino acid that has an *R* absolute configuration.

(Choice A) Although *most* natural amino acids have an *S* absolute configuration, cysteine has an *R* configuration.

(Choice B) The typical configuration of all chiral amino acids, including cysteine, is L configuration.

(Choice C) The α -carbon of glycine is achiral; therefore glycine has no stereochemical designation.

Educational objective:

The stereochemical designations of naturally occurring amino acids are typically L and S. The two exceptions are glycine, which is achiral, and cysteine, which is *R*, but still L.

Time spent:0 sec

Subject:
Biochemistry

QID:404175

System:
Amino Acids and Proteins

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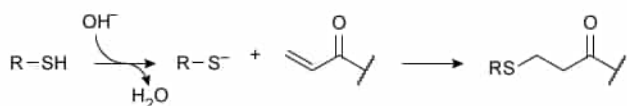


Figure 1 Reaction of a thiol with an acrylamide

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Experiment 2

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At a pH of 7.0, the G12C cysteine residues of mutant Ras proteins are, on

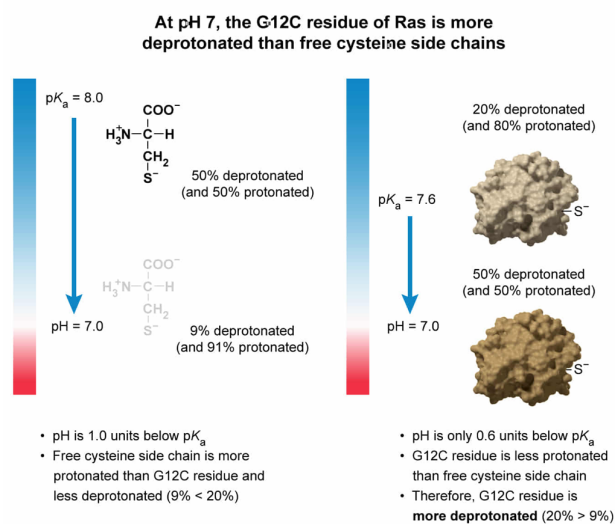
average:

- ✔ ☒ A. less protonated than free cysteine side chains. [42%]
- ☐ B. more protonated than free cysteine side chains. [27%]
- ☐ C. protonated to the same extent as free cysteine side chains. [26%]
- ☐ D. fully (100%) protonated. [3%]

Correct

42% answered correctly

Explanation:



In solution, **ionizable functional groups** exist in an equilibrium between their protonated and deprotonated forms. The pK_a of a functional group is a measure of its acidity and is equal to the pH at which the protonated and deprotonated forms are present at **equal concentrations**. As the pH rises above the pK_a (ie, protons are removed from solution), the protonated form of the molecule decreases, and the deprotonated form predominates. As the **pH drops below the pK_a** (ie, protons are added to the solution), the **protonated form of the molecule predominates** and the **deprotonated form decreases**.

The passage states that free cysteine residues have a side chain pK_a of 8.0. Therefore, a pH of 7.0 is 1.0 pH unit below the pK_a . In contrast the **G12C** residue of mutant Ras has a side chain pK_a of 7.6. A pH of 7.0 is **only 0.6 pH units below the pK_a** . Therefore, the equilibrium of the G12C residue is **less shifted toward the protonated form**, which means it is **more shifted towards its deprotonated form**.

This shift of protonation state equilibrium is a common feature of enzymes that enhances the nucleophilicity of **particular amino acids**. In addition, the protonation–deprotonation equilibria can also be quantitatively calculated using the **Henderson–Hasselbalch equation**. The Henderson–Hasselbalch equation confirms that the **G12C residue of mutant Ras is more deprotonated than free cysteine side chains at pH 7.0**.

(Choice B) The G12C residue of mutant Ras is less protonated (more deprotonated) than free cysteines at pH 7.0.

(Choice C) Because the free cysteine side chain and the Ras mutant cysteine side chain have different pK_a values, they have different protonation equilibria and will therefore be protonated to different extents at a pH of 7.0.

(Choice D) Although both free cysteine and the Ras mutant cysteine are *predominantly* protonated at a pH of 7.0, neither are *fully* protonated. The extent of protonation **changes gradually** over a range of pH values and equals 50% when $pH = pK_a$.

Educational objective:

The pK_a value is a measure of acidity of a functional group. When $pH = pK_a$, the protonated and deprotonated forms of a functional group are present at equal concentrations. The concentration of the protonated form increases as the pH drops.

Time spent:0 sec

Subject:
Biochemistry

QID:404176

System:
Amino Acids and Proteins

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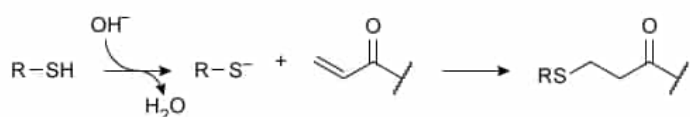


Figure 1 Reaction of a thiol with an acrylamide

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Experiment 2

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over time upon addition of ARS-853 was measured. Two time constants were calculated from the fluorescent decay signal, consistent with a two-step reaction mechanism of binding followed by covalent attack.

Given that fluorescence of organic molecules is most common in molecules that show a high degree of pi bond conjugation, which of the following amino acids was most likely introduced into the Ras protein for Experiment 2?

- ☐ A. Serine [2%]
- ☐ B. Proline [3%]
- ☐ C. Lysine [2%]
- ✓ ☒ D. Tryptophan [92%]

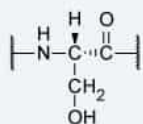
Correct

92% answered correctly

Explanation:

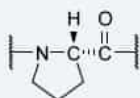
Tryptophan is an aromatic amino acid with extensive conjugation that allows for fluorescence

Serine



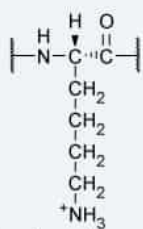
- Polar, nucleophilic amino acid
- No pi orbitals or sp^2 atoms in side chain, so no conjugation

Proline



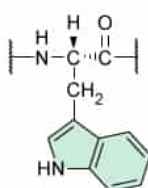
- Hydrophobic side chain connects back to backbone N
- No pi orbitals or sp^2 atoms in side chain, so no conjugation

Lysine



- Basic amino acid
- No pi orbitals or sp^2 atoms in side chain, so no conjugation

Tryptophan



- Aromatic amino acid
- Extensive conjugation across two rings allows for fluorescence

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Amino acids, the building blocks of proteins, exhibit different properties

based upon the functional groups and structural characteristics of their **side chains**. For example, ionizable functional groups confer charge and acid–base reactivity, hydroxyl groups confer polarity and nucleophilic potential, the proline side chain confers structural rigidity, and the glycine side chain confers flexibility.

The passage states that an amino acid was added to Ras_{G12C} that allowed Ras_{G12C} **fluorescence** intensity to be measured under different conditions. The question states that fluorescence in organic molecules can occur when a molecule has a high degree of **pi bond conjugation**. Pi conjugation occurs when a molecule has series of adjacent **pi bonds** (eg, double bonds) or *p* orbitals, which allows electrons within those orbitals to be delocalized across the conjugated system. This delocalization lowers both the ground and the excited-state energy levels of those electrons, allowing molecules to absorb ultraviolet light and fluoresce.

Of the amino acids presented, only **tryptophan** has conjugated pi bonds in its side chain. As the largest **aromatic amino acid** tryptophan has the largest conjugated pi bond system among the 20 amino acids, and therefore most readily absorbs ultraviolet light. This feature of tryptophan can be used for protein quantitation (by measuring absorbance at 280 nm) or **for measuring fluorescence**, as described in the passage.

(Choice A) **Serine** has a polar, nucleophilic hydroxyl group in its side chain but does not have a conjugated pi bond system.

(Choice B) **Proline's** side chain connects back to the peptide backbone, which confers structural rigidity to the protein. Although proline's side chain forms a ring, it is not an aromatic ring and does not contain a conjugated pi bond system.

(Choice C) **Lysine** has an amine group in its side chain but has no double bonds and therefore no pi orbitals to form a conjugated pi bond system.

Educational objective:

The aromatic amino acids (eg, tryptophan) have side chains with extensive pi bond conjugation, which allows them the absorb ultraviolet

light and fluoresce.

Time spent:0 sec

Subject:
Biochemistry

QID:404177

System:
Amino Acids and Proteins

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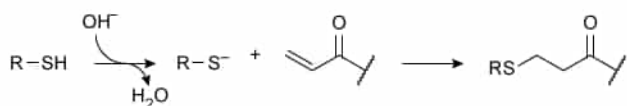


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The two time constants measured in Experiment 2 were τ_D and τ_I . These time

constants were pH dependent and pH independent, respectively. One of these constants represents noncovalent binding between ARS-853 and Ras_{G12C}, while the other represents the cysteine residue nucleophilically attacking ARS-853.

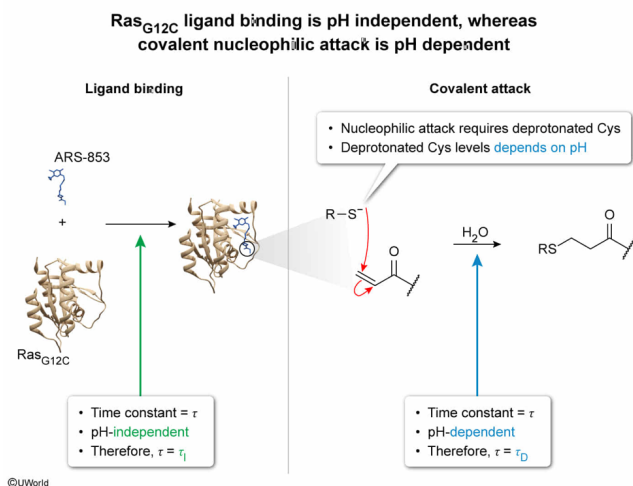
Based on the information in the passage, the most likely interpretation is that:

- ☐ A. τ_D represents the ligand binding reaction and τ_I represents the nucleophilic attack. [25%]
- ✔ ☒ B. τ_D represents the nucleophilic attack and τ_I represents the ligand binding reaction. [55%]
- ☐ C. The time constants represent ligand binding and nucleophilic attack but cannot be assigned without further information. [12%]
- ☐ D. the time constants cannot be specifically correlated to ligand binding or nucleophilic attack because the fluorescent amino acid is far away from the binding and attack site. [6%]

Correct

56% answered correctly

Explanation:



Many biochemical reactions occur over multiple steps. Typically, the first step is **binding** (ie, collision of the protein or enzyme with the other reactant, as well as the subsequent **conformational changes** leading to an **induced fit**). After binding, the catalytic step or steps comprise the atomic collisions that lead to formation and/or breaking of covalent bonds. Scientists can distinguish the different steps of a reaction by the kinetic and thermodynamic properties of those steps.

In the passage, two **time constants** were identified in Experiment 2, which correspond to the two steps of the overall reaction: binding and catalytic attack.

In the passage, Experiment 1 showed that the **binding step** (as measured through K_d) was **not significantly pH dependent**. Therefore, the most likely interpretation is that **the binding step is represented by the pH-independent time constant τ_i** .

In contrast, the catalytic step involves nucleophilic attack by the cysteine side chain to the electrophilic alkene in the acrylamide group. Figure 1 shows that the **deprotonated thiolate**, rather than the protonated thiol, of cysteine acts as the nucleophile. The concentration of the deprotonated thiolate **depends on the pH**; therefore, **the rate of the catalytic step is most likely pH dependent and represented by the time constant τ_D** .

In summary, the pH-dependent τ_D most likely represents the pH-dependent nucleophilic attack step, whereas the pH-independent τ_i most likely represents the pH-independent binding step.

(Choices A and C) The binding reaction is pH independent and represented by τ_i . The nucleophilic attack step is pH dependent and represented by τ_D .

(Choice D) Effects at the drug-binding site can be transmitted through conformational changes to other parts of the protein, including a faraway fluorescent amino acid. This is an example of **protein allostery**.

Educational objective:

Different steps of a reaction mechanism can be distinguished by the properties of those individual steps. A reaction that depends on the protonation state of one of the reactants is likely to be dependent on pH.

Time spent:0 sec

Subject:
Biochemistry

QID:404178

System:
Amino Acids and Proteins